

Crucial: We are implicitly conditioning on LBW. see how we compare

Redo table with

Smoker Non-smoker

LBW	NBW
-	-
-	-

Smoker Non-smoker

LBW	NBW
-	-
-	-

Smoking mom & BW

if LBW, good mom smokes

8.4 A more complex, but very real, example: the birth-weight paradox

The “birth weight paradox” is explained in detail in Hernandez-Diaz *et al.* (2006). The Wikipedia has it under the name “Low birth-weight paradox” (https://en.wikipedia.org/wiki/Low_birth-weight_paradox). This is an issue of major practical relevance that, if poorly analyzed and understood, could lead to harmful recommendations.

I copy from p. 1115 of Hernandez-Diaz *et al.* (2006):

① “Low birth weight (LBW) infants in groups with a high prevalence of LBW have a lower mortality rate than LBW infants in groups with a low prevalence of LBW, whereas the opposite is true for normal-weight infants. (...)”

② For example, when studies compared mortality rates between LBW infants born to smokers and nonsmokers, infants of smokers had lower mortality rates.

③ Although it is widely accepted that infants born to mothers who smoke have lower birth weights and are at higher risk of neonatal mortality, it has been suggested that the effect of maternal smoking is modified by birth weight in such a way that smoking is beneficial for LBW babies.”

Conditioning variable

(Does the above sound confusing? It was for me. Write, with your own words and your own pencil, a simple scheme of what is being described.)

This is another case of collider bias, in this case stratifying on a common effect. In Figure 12 there are two possible causal DAGs that would lead to this paradox.

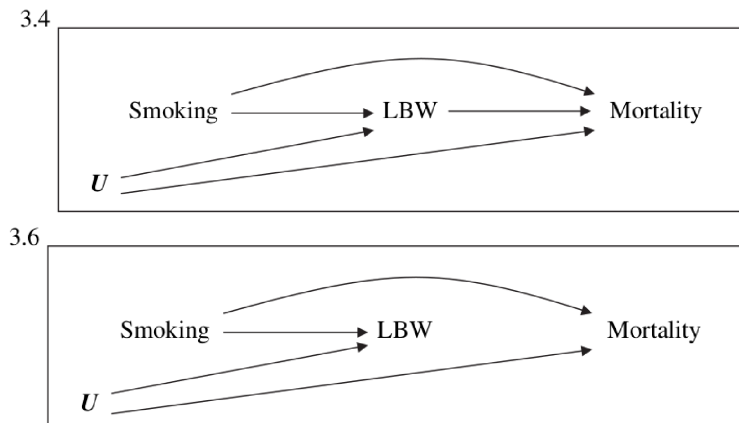


Figure 12: . Two DAGs that would result in the birth-weight paradox. From Figure 3 in Hernandez-Diaz *et al.* (2006). The presence or not of an arrow from LBW to Mortality is not the relevant feature.

And what is U and what is an intuitive explanation of this phenomenon? Low birth weight can be caused by medical conditions other than smoking. Once we know a baby has low birth weight, if we know it is the child of a smoking mom, then it is less likely that the baby has those other conditions (this is the usual “explaining away” of colliders; see the section where we discussed colliders, Berkson’s paradox, and dating men who are handsome jerks —section 6). In fact, we do not need LBW to have any effect on mortality. Look at the last DAG in Figure 12: Smoking is harmful, but (as explained in the Wikipedia) other causes of low birth weight can be more harmful

The conditioning variable.

than smoking. Thus, among LBW babies, those from smoking moms show lower death rates. But telling pregnant women to smoke would be a very bad idea.

Again, the moral of the story is: do not adjust for covariates just because they have been measured (Hernández-Díaz *et al.*, 2006).

9 How is all of this relevant if I only do experimental/observational work?

9.1 Relevance

If your work involves observational data, and you want to interpret the estimates from your analyses as measures of effect, this is obviously relevant (this is what all the examples here show ☺).

If you only do experimental work, often you will want to adjust for additional covariates and, then, knowing what covariates to use becomes crucial if you want to correctly interpret the effects from your experimental manipulations. Moreover, causal inference from experimental data arguably makes use of counterfactuals (see “*Randomization, exchangeability, counterfactuals, and potential outcomes*” (section 14)).

And regardless of whether you work with observational or experimental data, the causal framework is the appropriate framework to think about confounding, as we have seen above. This cannot be emphasized strongly enough, and if you have doubts, look at the example in “*A pretreatment collider variable we should not adjust on*” (section 8.1).

9.2 But I need to know at least something about the phenomenon I am dealing with to do any of this!!!!

Well, yes, of course you do. As Miguel Hernán emphasized in his edX course “Causal Diagrams: Draw Your Assumptions Before Your Conclusions”

<https://www.edx.org/course/causal-diagrams-draw-your-assumptions-before-your>, you have to draw your assumptions before your conclusions.

Causal claims are much stronger than claims about association. They require making stronger assumptions.

(By the way, we have not touched at all here on topics such as trying to measure the causes of an effect and finding **sufficient causes**, nor on trying to find the causal DAGs from a set of data. We have focused “just” on measuring the effects of causes. For more on those topics, go to Hernán and Robins, 2020; Morgan and Winship, 2015; Neal, 2020, Peters *et al.*, 2017).
